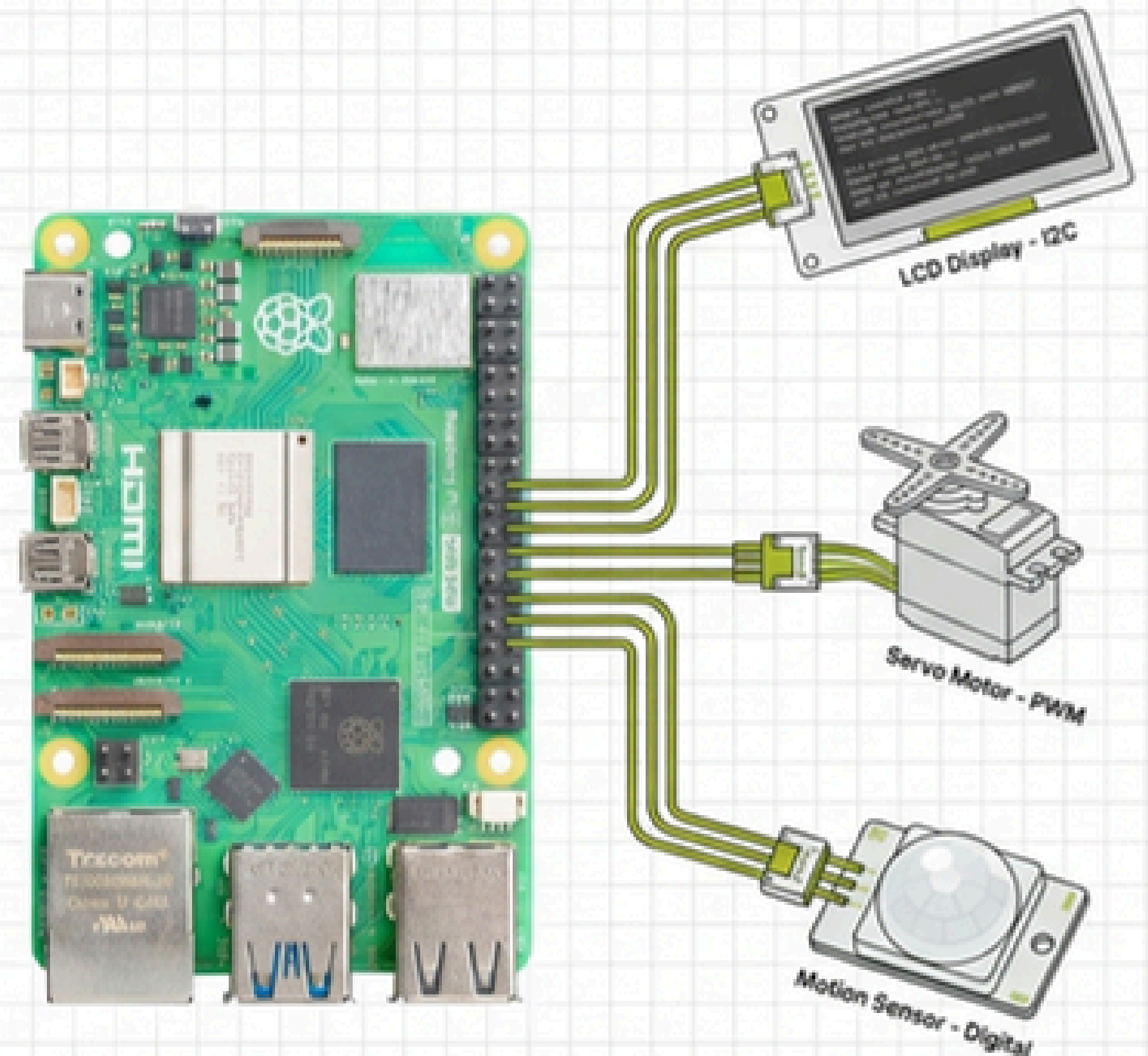
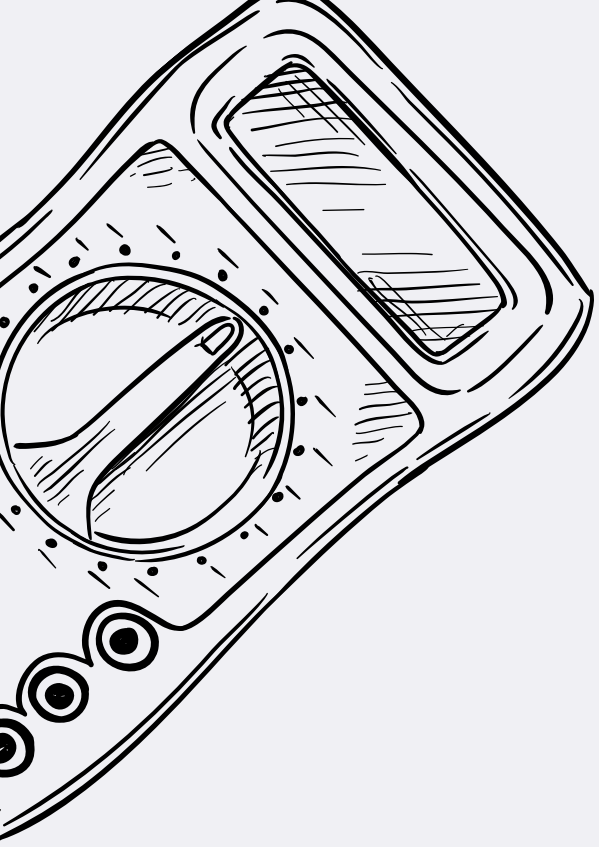


Constructing the Physical Interface

Workshop II: From Processing Core to Edge Interaction

- **Part 1:** The Foundation – Architecture & Ecosystem
- **Part 2:** The Toolkit – 7 Modular Components
- **Goal:** Mastering grove.py for Edge Devices

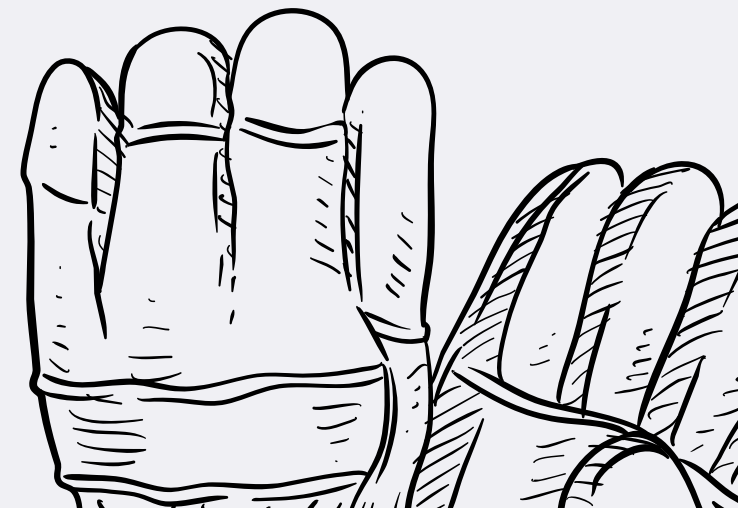
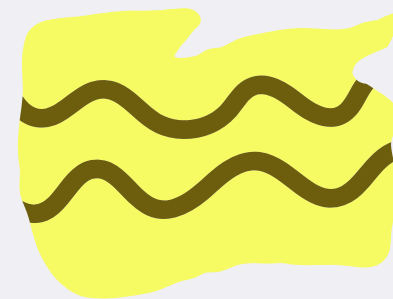
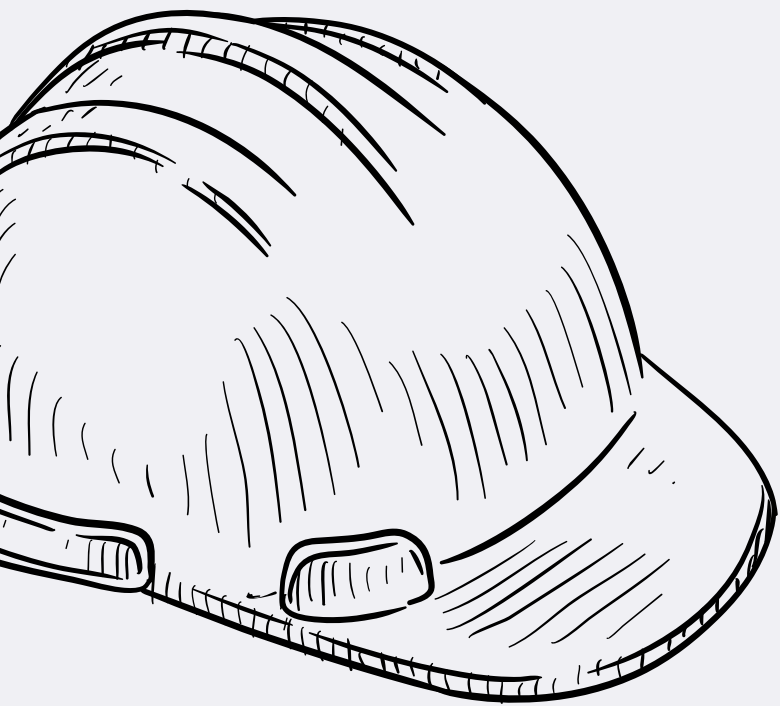
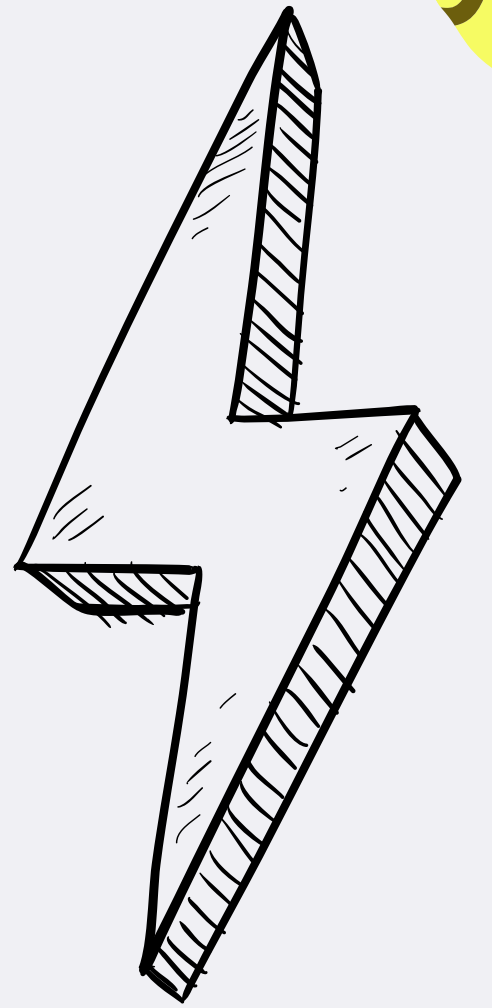




Brought to you by:

CIRCUIT REVIVAL

Delivered by Yuvraj



The Core: Raspberry Pi 5 Architecture

Connectivity

Dual-band 802.11ac Wi-Fi + Gigabit PoE+



Broadcom BCM2712 SoC

Quad-core Arm Cortex-A76 @ 2.4GHz

RP1 Southbridge

Dedicated I/O & Peripheral Management

Technical Data

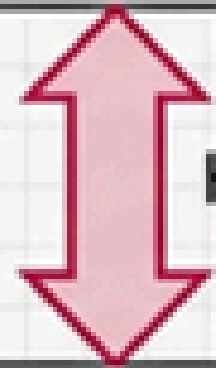
PERFORMANCE PROFILE:

- 2-3× CPU increase over Pi 4
- VideoCore VII GPU
- High-speed PCIe 2.0 Interface

The Environment: OS and Libraries

Python Interface

grove.py



Hardware Abstraction

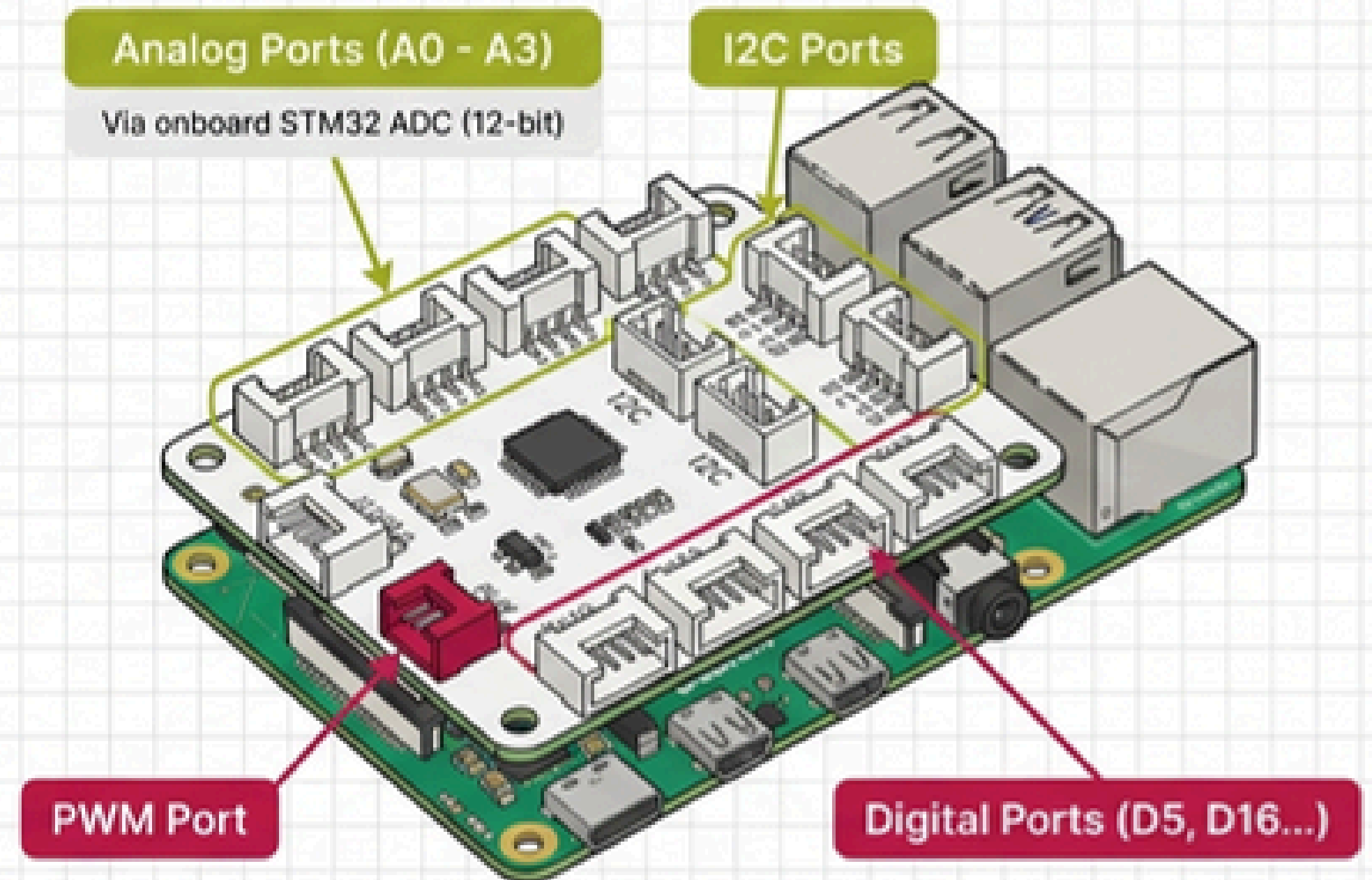
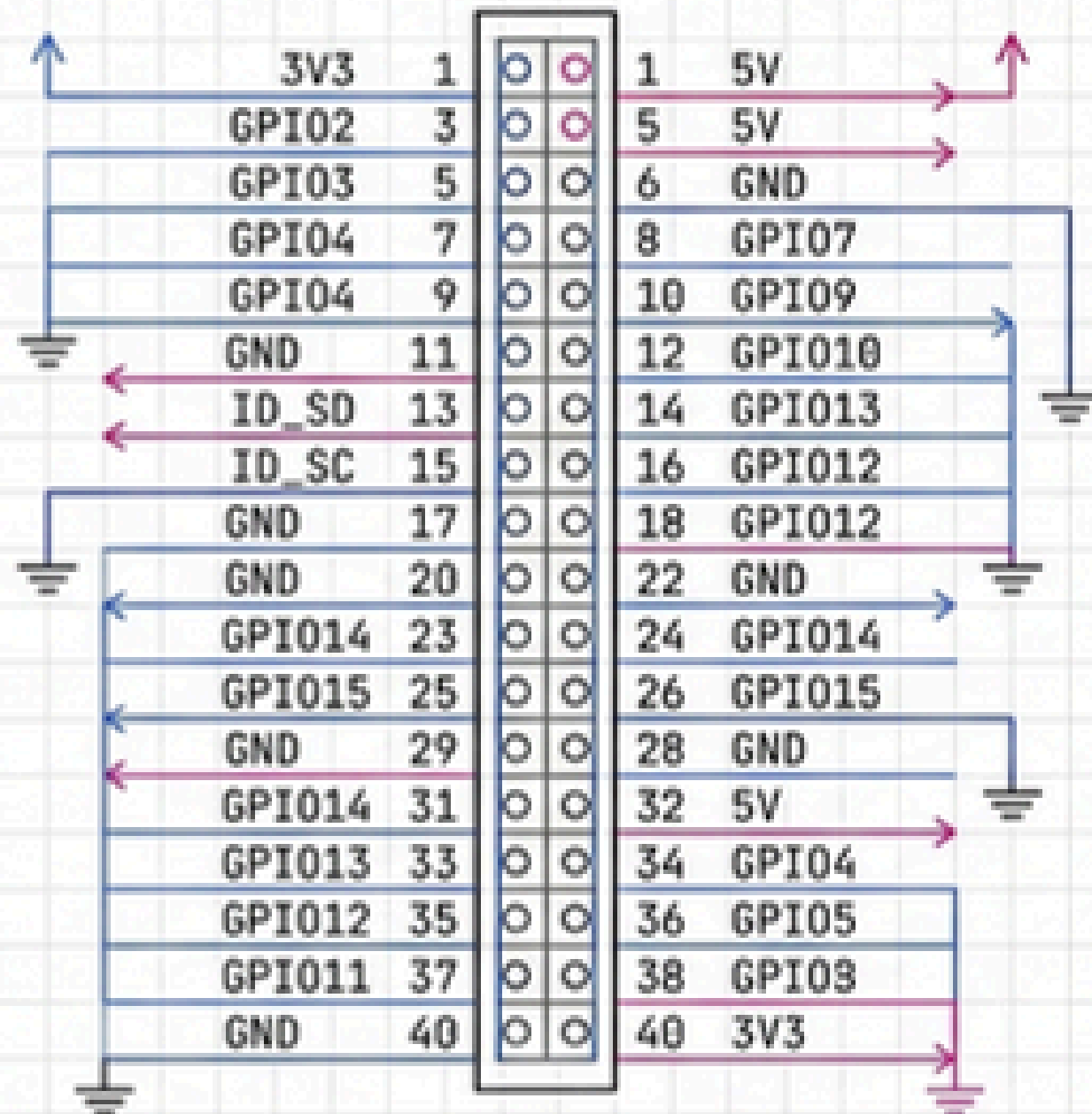
Raspberry Pi OS (Debian Bookworm)

- Optimized for Low Memory
- Wayland / labwc compositor

```
pi@raspberrypi:~ $ git clone https://github.com/Seed-Studio/grove.py
```

The `grove.py` library treats complex hardware behaviors as simple Python objects.

The Bridge: Grove Base Hat & GPIO



Critical Concept

The Raspberry Pi is purely digital. The Grove Base Hat adds an Analog-to-Digital Converter (ADC), enabling the use of analog sensors like light and sound detectors.

GOOGLE DRIVE Folder Workshop

**Scan this QR Code to go to my Google Drive
Folder that has the Sheets & Codes - a lot of
help to help you code**



Visual Output: 16x2 LCD Display



Specs

Type: I2C Interface
Pins Used: SDA, SCL (2 wires)
Driver: JHD1802

Python Code

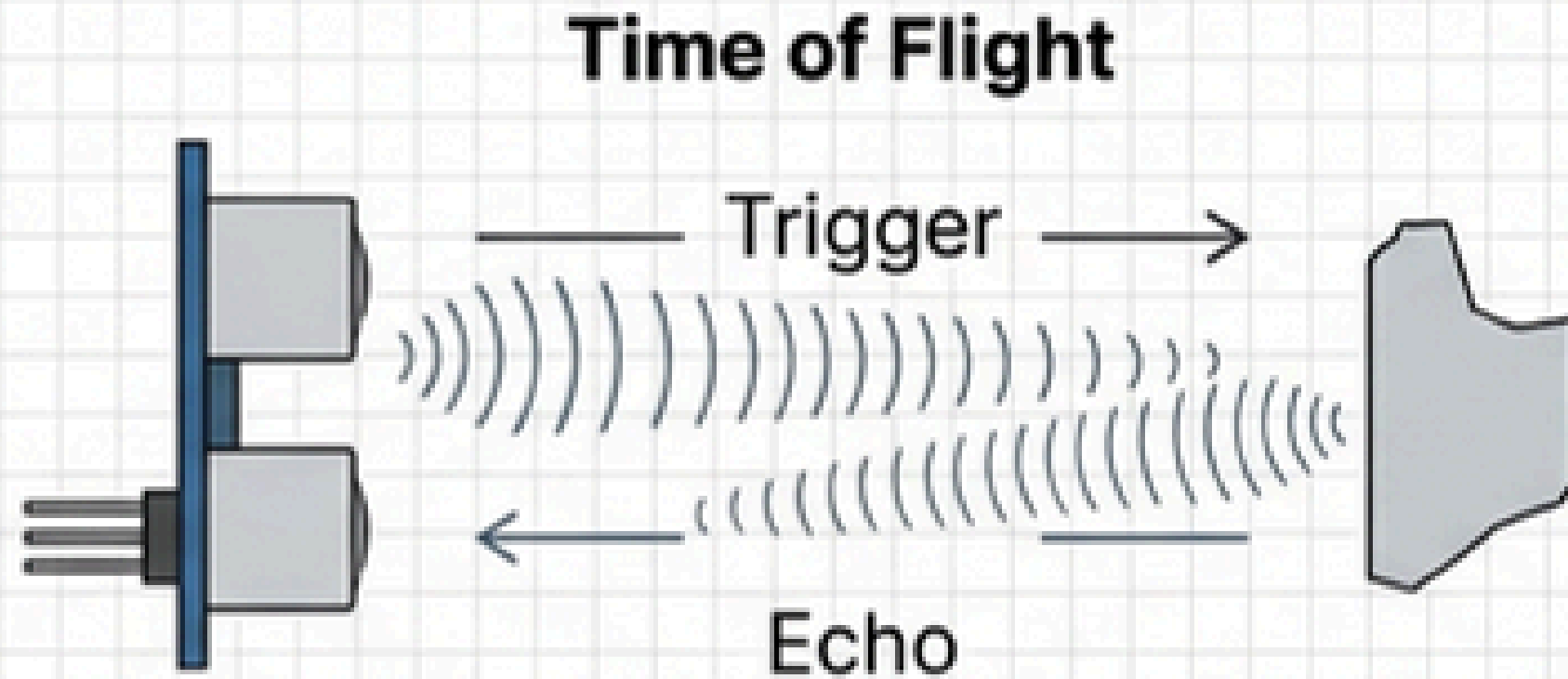
```
from grove.display.jhd1802 import JHD1802

lcd = JHD1802() # Auto-detects on I2C bus
lcd.setCursor(0, 0) # Column 0, Row 0
lcd.write('Hello World')
```

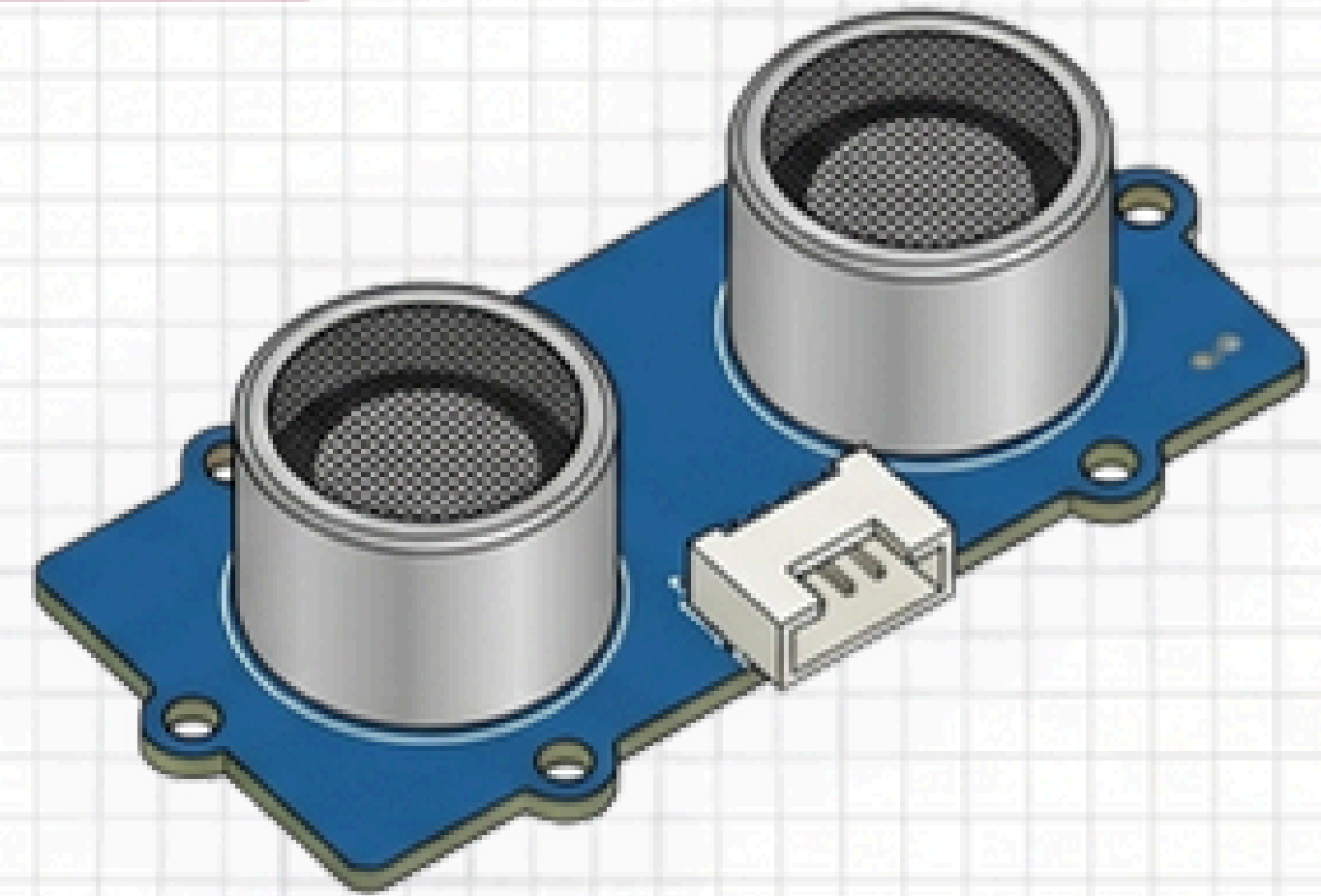
Grove Green

Why I2C? It allows complex data (text) to be sent over just 2 wires, leaving GPIO pins free for other sensors.

Spatial Sensing: Ultrasonic Ranger



$$\text{Distance} = \frac{(\text{Echo Time} \times \text{Speed of Sound})}{2}$$

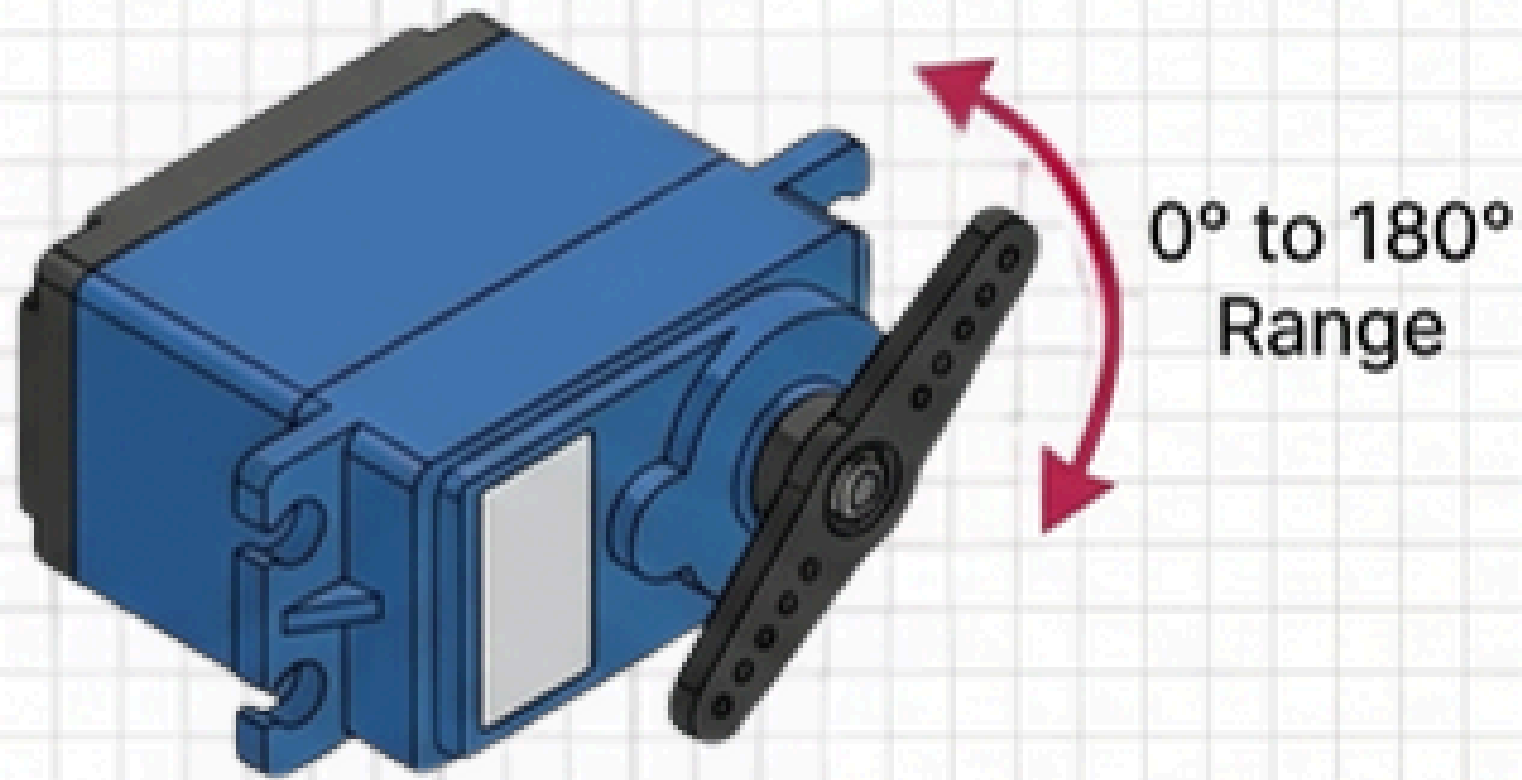


Range: 3cm – 350cm
Type: Digital GPIO

```
from grove.grove_ultrasonic_ranger import GroveUltrasonicRanger

# Connect to Digital Port D5
sensor = GroveUltrasonicRanger(5)
distance = sensor.get_distance()
print('{} cm'.format(distance))
```

Physical Actuation: Servo Motor



0° to 180°
Range



Specs

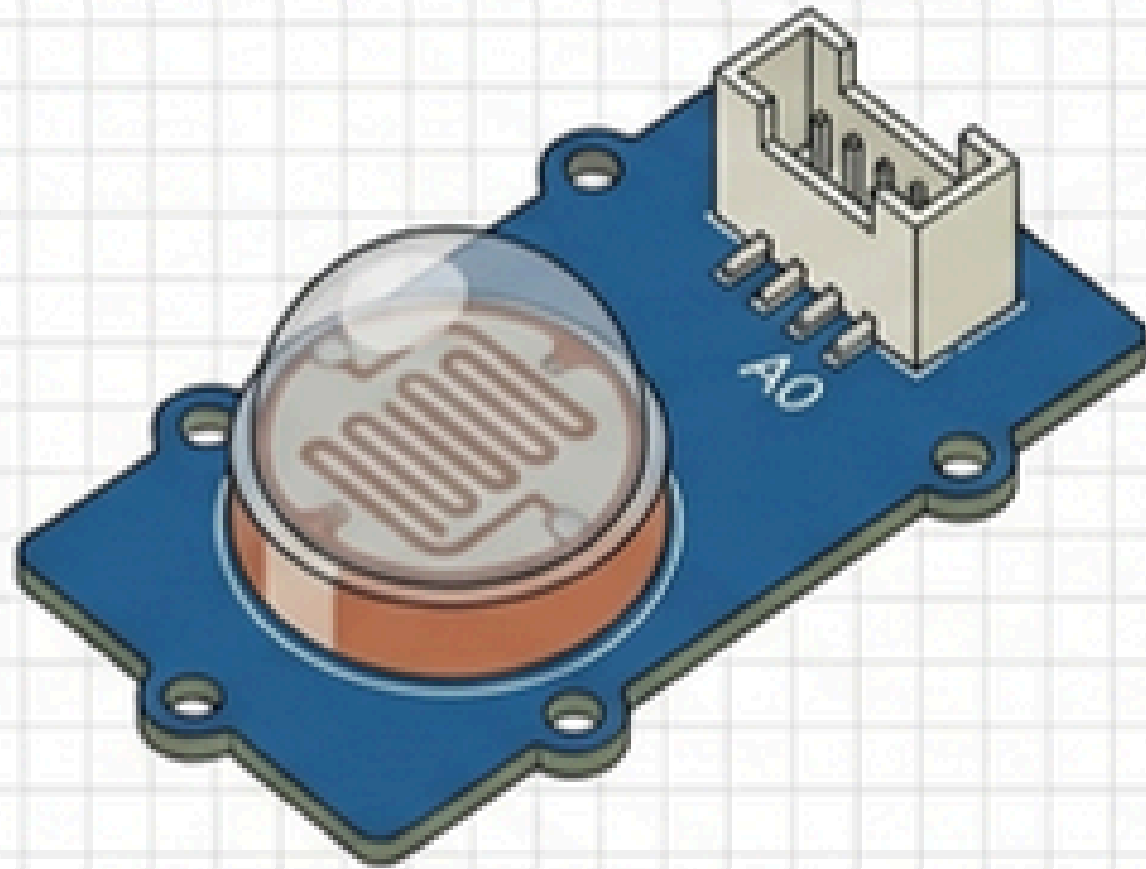
Type: PWM Interface
Application: Robotic Arms, Gimbals

Python Code

```
from grove.grove_servo import GroveServo

# Connect to PWM Port (Pin 12)
servo = GroveServo(12)
servo.setAngle(90) # Move to center
```


Environmental Sensing: Light Sensor



Must use Analog Port (A0)

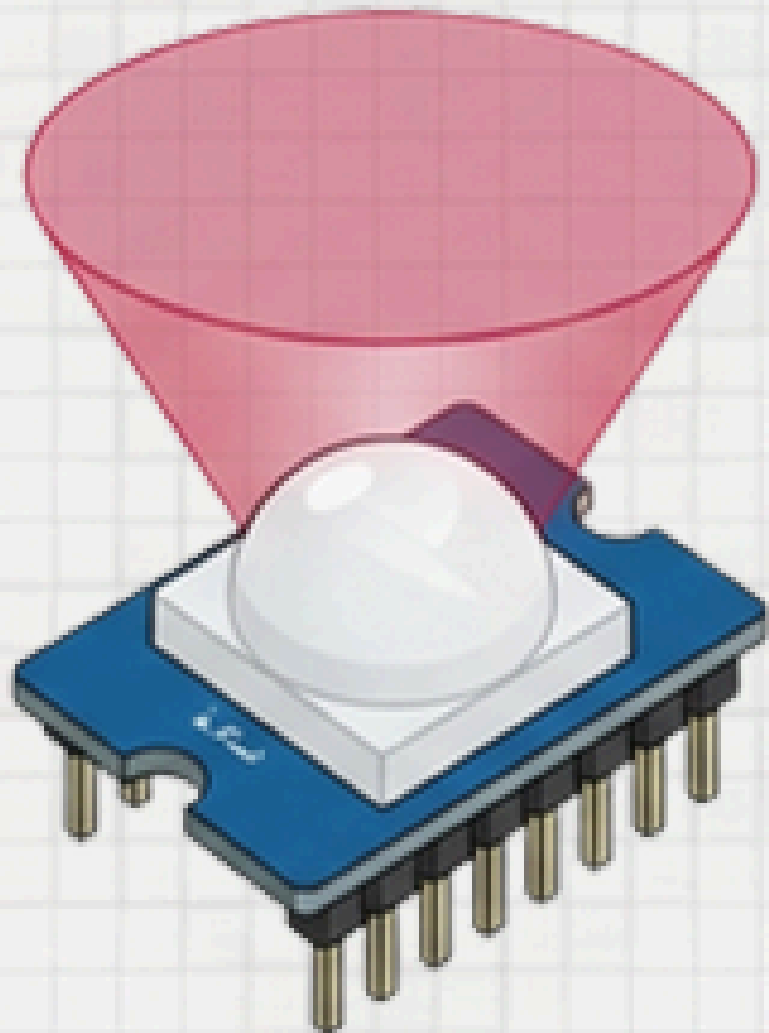
Physics: Resistance decreases as light intensity increases.

ADC Role: Base Hat converts voltage (0-3.3V) to integer.

Python Code

```
# Connect to Analog Port A0
sensor = GroveLightSensor(0)
print('Light value: {}'.format(sensor.light))
```

Presence Detection: Mini PIR Motion Sensor



Type: Digital GPIO
Mechanism: Passive
Infrared (Body
Heat Detection)
Logic: Binary (Motion /
No Motion)

Python Code

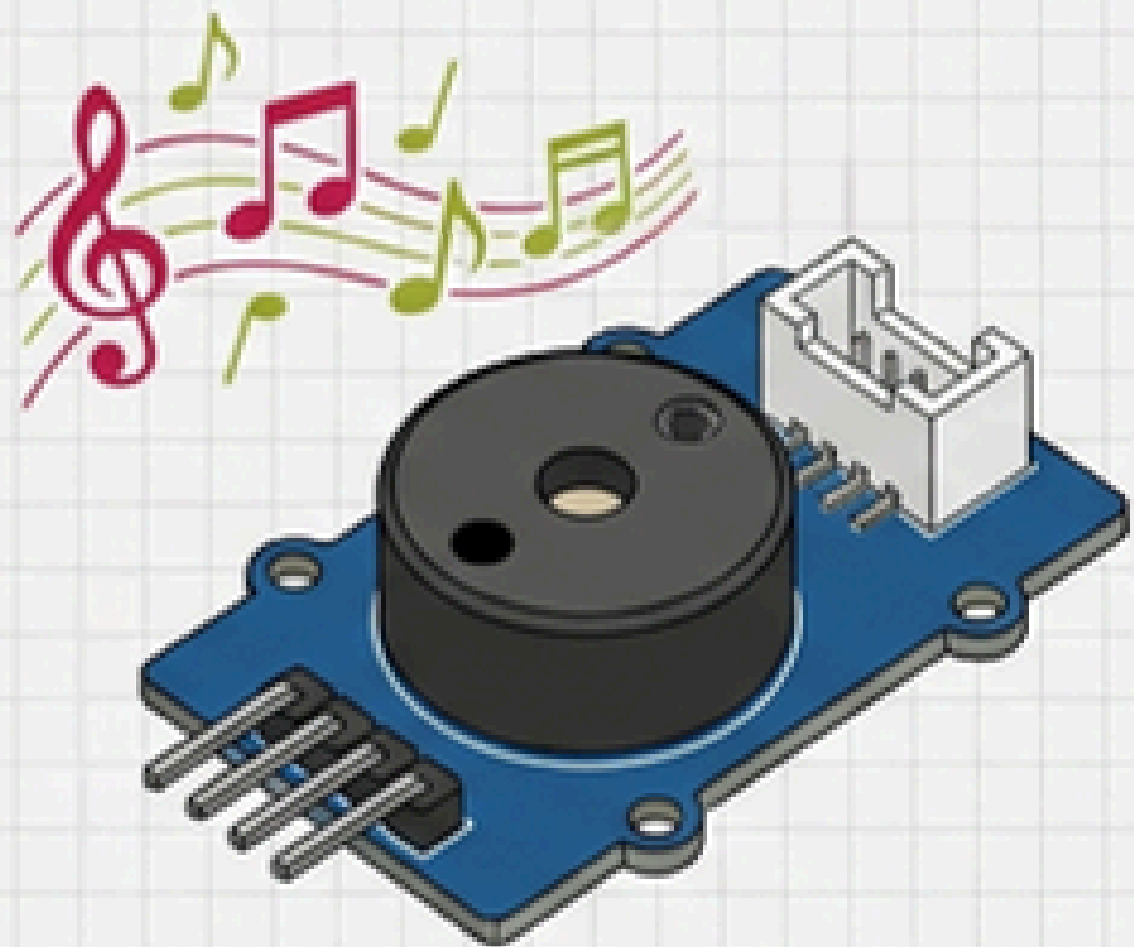
```
from grove.grove_mini_pir_motion_sensor
import GroveMiniPIRMotionSensor

# Connect to GPIO Port (Pin 5)
sensor = GroveMiniPIRMotionSensor(5)

def on_detect():
    print('Motion Detected!')

# Event Callback (No while loop needed)
sensor.on_detect = on_detect
```

Audio Output: Buzzer



Type: PWM / Digital

Modes: Simple Beep (Digital) or
Tone Generation (PWM Frequency)

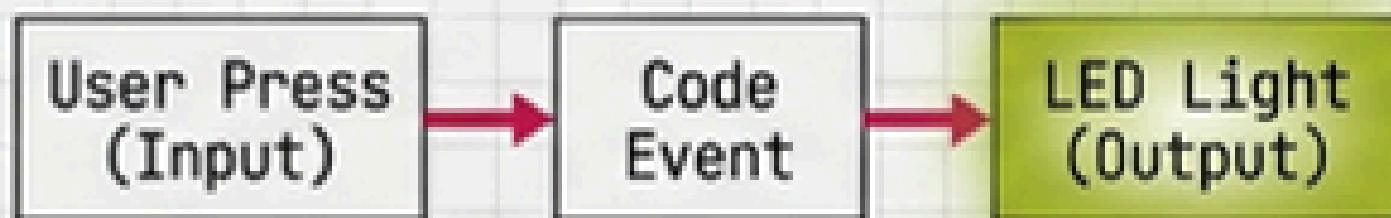
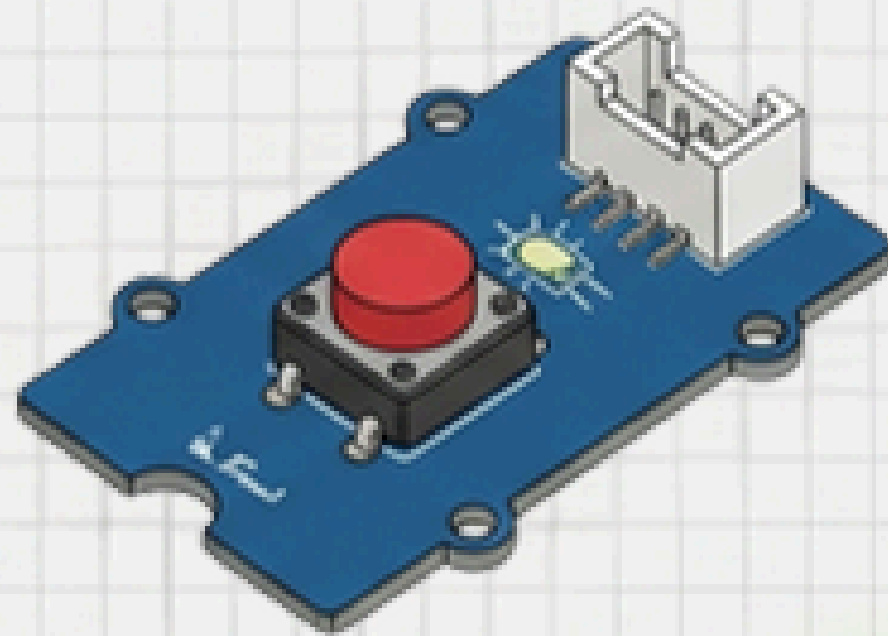
Python Code

```
from upm import pyupm_buzzer as upmBuzzer

# Connect to PWM Port (GPIO 12)
buzzer = upmBuzzer.Buzzer(12)

# Play 'Do' note for 0.5 seconds
buzzer.playSound(upmBuzzer.BUZZER_D0, 500000)
```


Interactive Input: Red Button & LED



Type: Digital GPIO (Composite)

Features: Momentary switch +
Integrated LED

Python Code

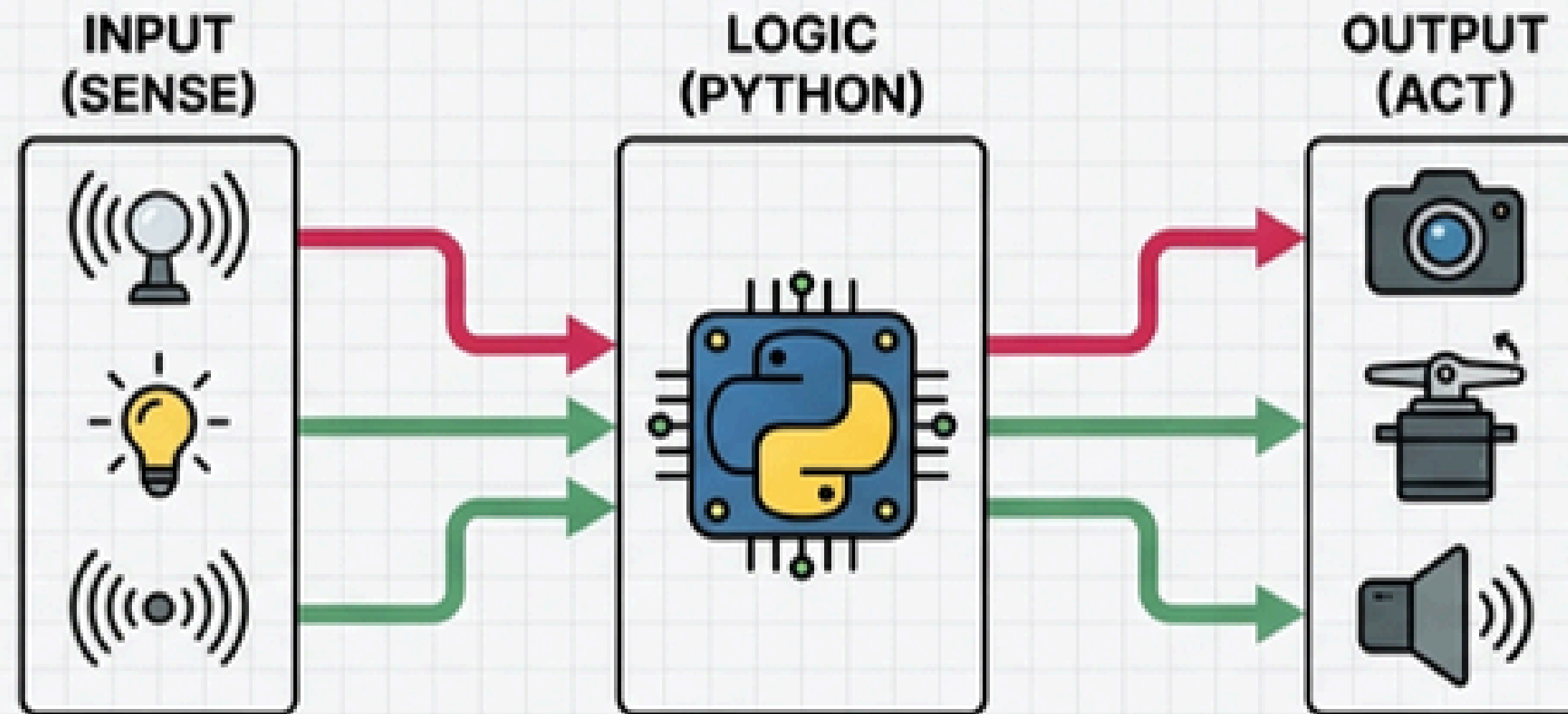
```
from grove.grove_ryb_led_button import GroveLedButton

button = GroveLedButton(5)

def on_event(index, event, tm):
    if event & GroveLedButton.EV_SINGLE_CLICK:
        button.led.light(True) # Feedback On
    if event & GroveLedButton.EV_LONG_PRESS:
        button.led.light(False) # Feedback Off

button.on_event = on_event
```

Synt. Synthesis: From Modules to Systems



1. **Smart Security:** PIR Motion (Input) → triggers Camera & Buzzer (Output)
2. **Environmental Control:** Light Sensor (Input) → triggers Servo Blinds (Output)
3. **Distance Monitor:** Ultrasonic Ranger (Input) → displays value on LCD (Output)

The hardware is ready. The logic is yours to write.

Red Light

Green Light

45.6 Billion Won
Is Child's Play

SQUID GAME



PROJECT 02: RED LIGHT - GREEN LIGHT TOURNAMENT

